Xiaozhi modifies the default prompt audio and converts its own P3 audio files (such as startup broadcast and network configuration broadcast)

1. Convert the audio to the specified sampling rate

You can use the following `ffmpeg` command to convert your own WAV file to a 16000 sample rate and set the sample depth to 32 bits:

bash

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```
ffmpeg -i input.wav -ar 16000 -acodec pcm_s32le output.wav
```

Parameter Description:

- `-i input.wav` : Input file.
- `-ar 16000` : Specifies the output sampling rate as 16000 Hz.
- `-acodec pcm_s32le` : Specifies the output audio encoding format as **32-bit little-endian PCM** (Linear PCM).
- `output.wav` : Output file name.

If higher quality resampling is required:

You can add the `-filter:a` option to use the `aresample` filter, specifying the quality of the resampling algorithm:

bash

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```
ffmpeg -i input.wav -ar 16000 -acodec pcm_s32le -filter:a "aresample=resampler=soxr" output.wav
```

This will use the `soxr` high-quality resampling algorithm.

2. Convert wav to P3 file

Copy the convert_audio_to_p3.py file from the Xia Ge source code directory to the directory where the audio files are located,

Run the following command:

```
pip install librosa opuslib tqdm numpy
```

Run the following command:

```
python .\convert_audio_to_p3.py output.wav output.p3
```

Place the output p3 file in the xiaozhi-esp32\main\assets\zh-CN directory, then recompile to see your own audio in lang_config.h

```
        extern const char p3_upgrade_start[] asm("_binary_upgrade_p3_sta
        extern const char p3_upgrade_end[] asm("_binary_upgrade_p3_end")
        static const std::string_view P3_UPGRADE {
        static_cast<const char*>(p3_upgrade_start),
        static_cast<size_t>(p3_upgrade_end - p3_upgrade_start)
        };

        extern const char p3_welcome_start[] asm("_binary_welcome_p3_sta
        extern const char p3_welcome_end[] asm("_binary_welcome_p3_end")
        static const std::string_view P3_WELCOME {
        static_cast<const char*>(p3_welcome_start),
        static_cast<size_t>(p3_welcome_end - p3_welcome_start)
        };

        extern const char p3_wificonfig_start[] asm("_binary_wificonfig_
        extern const char p3_wificonfig_end[] asm("_binary_wificonfig_p3
        static const std::string_view P3_WIFICONFIG {
        static_cast<const char*>(p3_wificonfig_start),
        static_cast<size_t>(p3_wificonfig_end - p3_wificonfig_start)
        };

        extern const char p3_wozai_start[] asm("_binary_wozai_p3_start")
        extern const char p3_wozai_end[] asm("_binary_wozai_p3_end");
        static const std::string_view P3_WOZAI {
        static_cast<const char*>(p3_wozai_start),
        static_cast<size_t>(p3_wozai_end - p3_wozai_start)
        };
    }
}
```

3. How to call the P3 file

For example, if I want to change the prompt tone of the network configuration, find WIFICONFIG

main > boards > common > wifi_board.cc > EnterWifiConfigMode()

board Esp32S3Korvo2V3Board 第 7 项,

```
WifiBoard::WifiBoard() {

std::string WifiBoard::GetBoardType() {
    return "wifi";
}

void WifiBoard::EnterWifiConfigMode() {
    auto& application = Application::GetInstance();
    application.SetDeviceState(kDeviceStateWifiConfiguring);

    auto& wifi_ap = WifiConfigurationAp::GetInstance();
    wifi_ap.SetLanguage(Lang::CODE);
    wifi_ap.SetSsidPrefix("Xiaozhi");
    wifi_ap.Start();

    // 显示 WiFi 配置 AP 的 SSID 和 Web 服务器 URL
    std::string hint = Lang::Strings::CONNECT_TO_HOTSPOT;
    hint += wifi_ap.GetSsid();
    hint += Lang::Strings::ACCESS_VIA_BROWSER;
    hint += wifi_ap.GetWebServerUrl();
    hint += "\n\n";

    // 播报配置 WiFi 的提示
    application.Alert(Lang::Strings::WIFI_CONFIG_MODE, hint.c_str(), "", Lang::Sounds::P3_WIFICONFIG);

    // Wait forever until reset after configuration
    while (true) {
        int free_sram = heap_caps_get_free_size(MALLOC_CAP_INTERNAL);
        int min_free_sram = heap_caps_get_minimum_free_size(MALLOC_CAP_INTERNAL);
        ESP_LOGI(TAG, "Free internal: %u minimal internal: %u", free_sram, min_free_sram);
        vTaskDelay(pdMS_TO_TICKS(10000));
    }
}

void WifiBoard::StartNetwork() {
    // User can press BOOT button while starting to enter WiFi configuration mode
```

Change it to your own; mine is WOZAI

```
    return "wifi";
}

void WifiBoard::EnterWifiConfigMode() {
    auto& application = Application::GetInstance();
    application.SetDeviceState(kDeviceStateWifiConfiguring);

    auto& wifi_ap = WifiConfigurationAp::GetInstance();
    wifi_ap.SetLanguage(Lang::CODE);
    wifi_ap.SetSsidPrefix("Xiaozhi");
    wifi_ap.Start();

    // 显示 WiFi 配置 AP 的 SSID 和 Web 服务器 URL
    std::string hint = Lang::Strings::CONNECT_TO_HOTSPOT;
    hint += wifi_ap.GetSsid();
    hint += Lang::Strings::ACCESS_VIA_BROWSER;
    hint += wifi_ap.GetWebServerUrl();
    hint += "\n\n";

    // 播报配置 WiFi 的提示
    application.Alert(Lang::Strings::WIFI_CONFIG_MODE, hint.c_str(), "", Lang::Sounds::P3_WOZAI);

    // Wait forever until reset after configuration
    while (true) {
        int free_sram = heap_caps_get_free_size(MALLOC_CAP_INTERNAL);
        int min_free_sram = heap_caps_get_minimum_free_size(MALLOC_CAP_INTERNAL);
        ESP_LOGI(TAG, "Free internal: %u minimal internal: %u", free_sram, min_free_sram);
        vTaskDelay(pdMS_TO_TICKS(10000));
    }
}
```

Just recompile and download